

NLP-Based Customer Query Intent Classification and Automatic Routing Using TF-IDF and Machine Learning

Mini NLP Project Report

Submitted By:

Name: **SONA DEVASIA**

Register Number: **24UBC155**

Class: **III BCA A**

Course: Natural Language Processing

1. Title

NLP-Based Customer Query Intent Classification and Automatic Routing Using TF-IDF and Machine Learning

2. Problem Statement

Customer-support systems receive a large number of text-based queries related to bookings, travel, transportation, weather, restaurants, and other services. Manually identifying the purpose of each query can be time-consuming.

This project develops an NLP-based system that automatically identifies the **intent** of a customer query and suggests an appropriate support route.

The system classifies queries into ten selected intents:

- Book Flight
- Book Hotel
- Flight Status
- Lost Luggage

- Restaurant Reservation
- Restaurant Suggestion
- Weather
- Car Rental
- Travel Suggestion
- Directions

For example, a query such as *"I want to book a flight to Delhi"* can be classified as book_flight and routed to **Flight Booking Support**.

The project demonstrates how NLP and machine learning can be used to automate customer-query classification and routing.

3. Problem Relevance

Customer queries are written in natural language and can express the same intention using different words and sentence structures. Therefore, NLP is suitable for identifying patterns in these queries.

The system can be useful in:

- Customer-support applications
- Travel-service platforms
- Chatbots
- Help-desk systems
- Automated query-routing systems

The output of the system is the **predicted intent** and its corresponding **support route**.

4. Objectives

The main objectives are:

1. To develop a practical NLP-based customer-query classification system.
2. To use a publicly available intent-classification dataset.
3. To select relevant customer-support intents from the dataset.
4. To preprocess the text data.
5. To convert text into numerical features using TF-IDF.
6. To implement Logistic Regression and Multinomial Naive Bayes.
7. To compare the performance of both models.
8. To evaluate the models using accuracy, precision, recall, F1-score, and a confusion matrix.
9. To classify new customer queries.
10. To create a simple automatic routing mechanism.
11. To identify limitations and suggest future improvements.

5. Dataset

5.1 Dataset Name

The dataset used is **CLINC150**, an intent-classification dataset containing 150 in-domain intent categories.

5.2 Dataset Source

The dataset was obtained in JSON format as `data_full.json` from the CLINC150 dataset repository.

The original dataset was introduced in the research paper *An Evaluation Dataset for Intent Classification and Out-of-Scope Prediction* by Larson et al. (2019).

5.3 Dataset Size

The original in-domain dataset contains:

<i>Split</i>	<i>Samples</i>
<i>Training</i>	15,000
<i>Validation</i>	3,000
<i>Testing</i>	4,500

Each intent contains 100 training, 20 validation, and 30 test examples.

5.4 Selected Dataset

For this project, ten relevant intents were selected from the original 150 intents.

<i>No.</i>	<i>Intent</i>
<i>1</i>	book_flight
<i>2</i>	book_hotel
<i>3</i>	flight_status
<i>4</i>	lost_luggage
<i>5</i>	restaurant_reservation
<i>6</i>	restaurant_suggestion
<i>7</i>	weather
<i>8</i>	car_rental
<i>9</i>	travel_suggestion
<i>10</i>	directions

The final dataset used in the experiment contained:

<i>Split</i>	<i>Samples</i>
<i>Training</i>	1,000

<i>Validation</i>	200
<i>Testing</i>	300
<i>Total</i>	1,500

5.5 Attributes

The main fields used were:

<i>Attribute</i>	<i>Description</i>
<i>text</i>	Customer query
<i>intent</i>	Target intent/category

Therefore:

Input: Customer query text

5.6 Data Preparation

The dataset was prepared by:

1. Loading the JSON dataset.
2. Converting the data into Pandas DataFrames.
3. Exploring the available intents.
4. Selecting ten relevant intents.
5. Filtering the training, validation and test sets.
6. Separating text and intent labels.
7. Preprocessing the text.
8. Converting the cleaned text into TF-IDF features.

6. Methodology

The project follows the pipeline:

Customer Query → Preprocessing → TF-IDF → Machine-Learning Model → Predicted Intent → Automatic Routing

6.1 Text Preprocessing

The customer queries were cleaned before classification.

The preprocessing includes:

- Converting text to lowercase
- Removing punctuation
- Removing unnecessary characters
- Removing extra spaces

For example:

Original:

I WANT to BOOK a Flight!!! Can you help me?

After preprocessing:

i want to book a flight can you help me

This reduces unnecessary variation in the text and prepares it for feature extraction.

6.2 TF-IDF

TF-IDF stands for **Term Frequency-Inverse Document Frequency**.

It converts text into numerical values based on the importance of words in the dataset.

Both single words and two-word combinations were considered using **unigrams and bigrams**.

The TF-IDF representation produced **4,307 features** for the final dataset.

6.3 Machine-Learning Models

Two models were implemented:

Logistic Regression

Logistic Regression was used for multi-class intent classification. It was selected because it works effectively with TF-IDF features and is simple to understand and implement.

Multinomial Naive Bayes

Multinomial Naive Bayes was used as a second classification model and baseline. It is commonly used for text classification and works well with numerical text features.

The assignment encourages simple, well-understood NLP solutions rather than unnecessarily complex models.

7. Implementation

7.1 Programming Language

The project was implemented using **Python** in **Google Colab**.

7.2 Libraries Used

<i>Library</i>	<i>Purpose</i>
<i>Pandas</i>	Dataset handling
<i>NumPy</i>	Numerical operations
<i>Scikit-learn</i>	TF-IDF, models and evaluation
<i>Matplotlib</i>	Visualization
<i>Seaborn</i>	Confusion matrix visualization
<i>re</i>	Text preprocessing
<i>JSON</i>	Dataset loading

7.3 Implementation Steps

The implementation consists of:

1. Dataset loading
2. Dataset exploration
3. Intent selection
4. Data preparation
5. Text preprocessing
6. TF-IDF feature extraction
7. Logistic Regression training
8. Naive Bayes training
9. Model comparison
10. Evaluation
11. Custom query prediction
12. Automatic query routing

8. Results

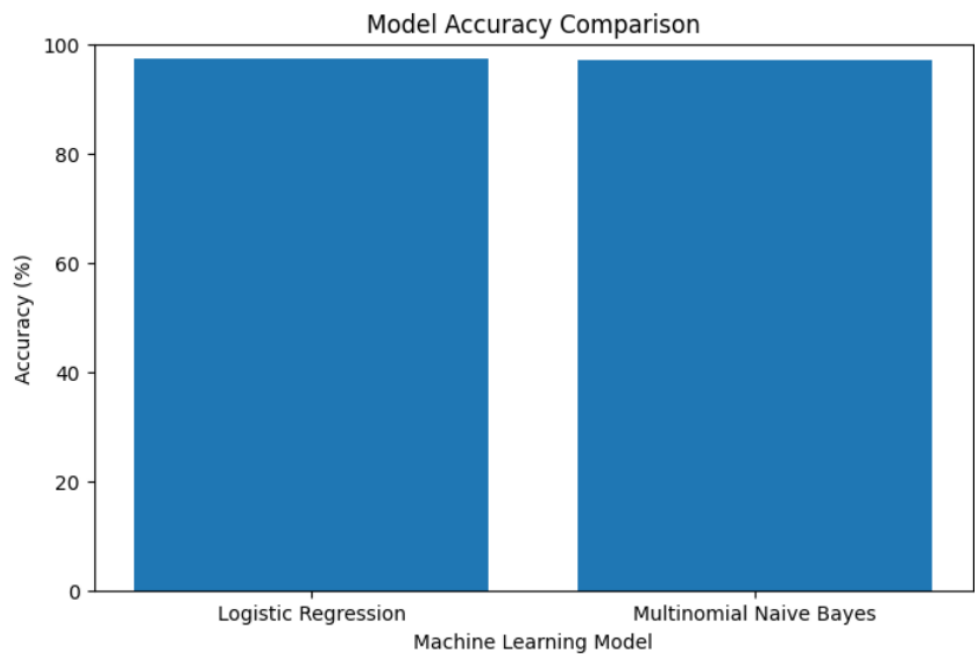
8.1 Model Comparison

The two models were evaluated on the 300 test samples.

<i>Model</i>	<i>Test Accuracy</i>
<i>Logistic Regression</i>	97.33%
<i>Multinomial Naive Bayes</i>	97.00%

Logistic Regression achieved the highest accuracy and was therefore used for the final prediction and routing demonstration.

Figure 1: Model Accuracy Comparison



8.2 Logistic Regression Performance

The Logistic Regression model achieved an overall test accuracy of **97.33%**.

The classification results were:

<i>Metric</i>	<i>Score</i>
<i>Precision</i>	0.97
<i>Recall</i>	0.97
<i>F1-score</i>	0.97
<i>Accuracy</i>	97.33%

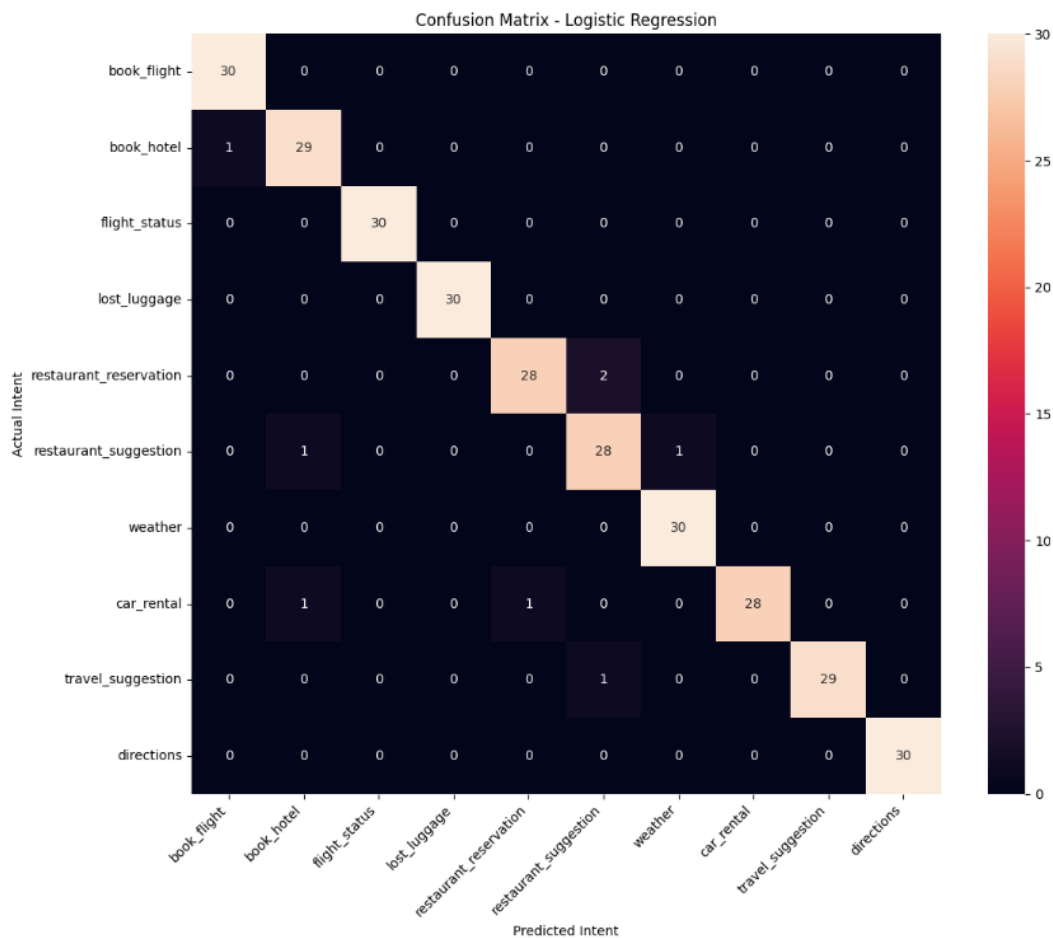
The results show that the model performed well across the selected ten intents.

8.3 Confusion Matrix

A confusion matrix was generated to analyze the correct and incorrect predictions of the Logistic Regression model.

Most predictions were concentrated along the diagonal, indicating that the majority of test queries were correctly classified.

Figure 2: Confusion Matrix



Some errors occurred between intents with similar vocabulary, particularly travel-related categories.

8.4 Sample Predictions

The trained model was tested using new customer queries.

<i>Customer Query</i>	<i>Predicted Intent</i>
<i>I want to book a flight to Delhi</i>	book_flight
<i>My luggage is missing</i>	lost_luggage
<i>Can you help me reserve a hotel?</i>	book_hotel
<i>Is my flight delayed?</i>	flight_status
<i>What is the weather today?</i>	weather

The results demonstrate that the trained model can classify new customer queries.

9. Automatic Query Routing

A simple routing mechanism was created by mapping each predicted intent to a support service.

<i>Intent</i>	<i>Support Route</i>
<i>book_flight</i>	Flight Booking Support
<i>book_hotel</i>	Hotel Booking Support
<i>flight_status</i>	Flight Status Support
<i>lost_luggage</i>	Baggage Support
<i>restaurant_reservation</i>	Restaurant Reservation Support
<i>restaurant_suggestion</i>	Restaurant Recommendation
<i>weather</i>	Weather Information
<i>car_rental</i>	Car Rental Support
<i>travel_suggestion</i>	Travel Planning Support
<i>directions</i>	Navigation Support

The complete process is:

Customer Query → Preprocessing → TF-IDF → Logistic Regression → Intent → Support Route

This demonstrates how the classification model can be used as part of a simple customer-support system.

Figure 3: Final System Prediction and Automatic Routing Output

```
Customer Query: I need to book a flight
Predicted Intent: book_flight
Suggested Route: Flight Booking Support
-----
Customer Query: Can you help me reserve a hotel?
Predicted Intent: book_hotel
Suggested Route: Hotel Booking Support
-----
Customer Query: Is my flight delayed?
Predicted Intent: flight_status
Suggested Route: Flight Status Support
-----
Customer Query: My luggage is missing
Predicted Intent: lost_luggage
Suggested Route: Baggage Support
-----
Customer Query: I want to reserve a table at a restaurant
Predicted Intent: restaurant_reservation
Suggested Route: Restaurant Reservation Support
-----
Customer Query: Can you suggest a good restaurant?
Predicted Intent: restaurant_suggestion
Suggested Route: Restaurant Recommendation Service
-----
Customer Query: What is the weather today?
Predicted Intent: weather
Suggested Route: Weather Information Service
-----
Customer Query: I need to rent a car
Predicted Intent: car_rental
Suggested Route: Car Rental Support
-----
Customer Query: Suggest a good place to travel
Predicted Intent: restaurant_suggestion
Suggested Route: Restaurant Recommendation Service
-----
Customer Query: How do I get to the airport?
Predicted Intent: directions
Suggested Route: Navigation Support
-----
```

10. Limitations

The project has the following limitations:

1. **Limited number of intents:** Only ten of the original 150 CLINC150 intents were used.

2. **Limited dataset size:** Only 1,500 examples were used for the final experiment.
3. **Limited contextual understanding:** TF-IDF focuses mainly on word importance and does not deeply understand sentence context.
4. **Similar intents:** Some categories contain similar vocabulary, which can cause incorrect predictions.
5. **No out-of-scope detection:** Queries outside the ten selected categories may still be assigned to one of the available intents.
6. **Prototype routing:** The routing system is a predefined mapping and is not connected to real customer-support departments.

11. Future Scope

The project can be improved in the following ways:

1. Use more of the available CLINC150 intents.
2. Add out-of-scope query detection.
3. Use a larger and more diverse dataset.
4. Improve preprocessing using tokenization, lemmatization and spelling correction.
5. Compare the current models with SVM or neural-network models.
6. Explore advanced transformer models such as BERT.
7. Add multilingual support.
8. Develop a web or mobile interface.
9. Connect the routing system to real customer-support services.
10. Perform more extensive evaluation using additional datasets.

12. Conclusion

The project developed an NLP-based prototype for classifying customer queries into ten travel and customer-support intent categories and automatically routing them to the appropriate support service. The project used the CLINC150 dataset and performed text preprocessing followed by TF-IDF feature extraction using unigrams and bigrams. Two machine-learning algorithms, Multinomial Naive Bayes and Logistic Regression, were implemented and compared.

Multinomial Naive Bayes achieved **97.00% test accuracy**, while Logistic Regression achieved **97.33% test accuracy**. Therefore, **Logistic Regression was selected as the better-performing model**. It achieved a precision of **0.97**, recall of **0.97**, and F1-score of **0.97** on the 300 test queries.

The final system was also tested with new customer queries and successfully predicted their intents and mapped them to suitable support routes. The project demonstrates that traditional NLP techniques such as TF-IDF, combined with machine-learning classification, can effectively organize customer queries for a focused set of intents.

At the same time, the results show that queries with similar vocabulary can sometimes be difficult to distinguish. The system can be improved in the future by using a larger number of intents and training examples, out-of-scope detection, better preprocessing, multilingual support, and advanced contextual models such as BERT.